CareCompanion: Assistive Companion Mobile Application for Elderly and Disabled Users

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***Abstract*—Mobile applications intended to help older adults manage medication and summon help are typically evaluated on the legibility of the older adult's interface. We argue that a second property matters at least as much and is rarely examined: whether the system tells the truth when it fails. We present CareCompanion, an implemented Android system for caregiver-mediated elder care built on three design commitments. First, an *asymmetric dual-surface* architecture places all configuration on the caregiver's device and leaves the older adult a zero-configuration surface of seven screens; measured from the source, the elder surface has a median declared text size of 20 sp against 13 sp on the caregiver surface, and every primary control equals or exceeds 64 dp against a 48 dp platform minimum. Second, *out-of-band device binding* resolves an identity gap created by federated sign-in: a single-use, short-lived numeric code issued to the caregiver binds the older adult's device to a care profile, replacing phone-number matching, which federated identity does not supply. Third, *fail-honest alerting* requires the interface never to assert an outcome it has not observed; an emergency alert that was not delivered says so and degrades to a one-tap, user-mediated fallback rather than displaying success. A structured audit of the feature-complete system found 13 defects, of which 8 (62%) would have failed with no user-visible signal, including two features that were written, stored, and never delivered. We report the taxonomy, the mechanisms that address each class, and an explicit account of what remains unevaluated: no user study has yet been conducted.**

***Index Terms*—elder care, mobile health, medication adherence, caregiver-mediated systems, accessibility, dependability, Android, human-centred design.**

I. INTRODUCTION

India is experiencing rapid population ageing, creating growing demand for accessible and scalable elder-care support. The United Nations Population Fund projects that India's population aged 60 and over will grow by 134% between 2022 and 2050, and the population aged 80 and over by 279% [1]. Digital access, however, remains uneven across age groups: although mobile-phone use is widespread nationally, internet use declines substantially with age [2], and among older adults in rural India — particularly women — shared handsets and limited digital literacy are the norm rather than the exception [3], [4].

Medication non-adherence is a significant concern among older adults managing chronic conditions and polypharmacy. Randomised evidence suggests mobile reminder tools help but do not solve the problem: a trial of a drug-management application among older adults with polypharmacy reported improved adherence and reduced readmission [5], while a 2026 systematic review of reminder technologies for home-dwelling older citizens reports beneficial effects for some outcomes but emphasises that the overall evidence remains heterogeneous and that further high-quality research is required [6].

The usability literature is clear about what makes an interface hard for an older adult. Small text is repeatedly the single most-cited barrier; touch targets are too small for reduced dexterity; icons are misread; and navigation depth exceeds working memory [7], [8], [9]. Design guideline syntheses converge on enlarged type, enlarged targets, shallow navigation, and error tolerance [10], [11].

What this literature largely assumes, however, is that the older adult is the person configuring the system. In practice, care is dyadic. Quinn *et al.* found in a dyad usability study that most interaction occurred between caregiver and older adult rather than the older adult alone [12], and dyadic digital health modules are now designed explicitly around shared care [13]. If the caregiver is doing the configuring, then the older adult's surface does not need to be a simplified version of a management application. It can be a different application altogether.

A second assumption is more consequential and, we argue, under-examined: that when the system reports success, it has succeeded. Emergency and reminder paths on mobile platforms are unreliable in ways the application can observe but frequently does not check. Android's Doze and App Standby defer alarms and network access during idle [14], and original-equipment-manufacturer power management is more aggressive still. Push delivery depends on connectivity that an older adult at home may not have. An interface that renders a green tick on an emergency alert it never delivered is not merely unhelpful; in the one situation the system exists for, it is actively harmful.

Contributions. This paper makes four contributions:

- An *asymmetric dual-surface* architecture for caregiver-mediated elder care, in which configuration authority and interface complexity are deliberately unequal across two clients over one care record, with the asymmetry measured from the implementation rather than asserted (Section III-A, Section V-A).
- *Out-of-band device binding*, a single-use short-lived code that binds a device to a care profile. We show that migrating to federated identity silently breaks phone-number-based linking, and that an out-of-band code is a simple, auditable repair (Section III-B).
- *Fail-honest alerting*: a discipline in which the interface never asserts an unobserved outcome, together with a permission-minimal degradation path that keeps the application distributable under current application-store policy (Section III-C, III-E).

- An empirical *defect taxonomy* from a structured audit of the feature-complete system, showing that silent failures — including two entire features that were stored but never delivered — dominate (Section V-C).

We are explicit about scope. This is a systems and design-contribution paper. We report conformance, interaction cost, and defect data measured from the artefact. We have not yet run a user study, and we make no usability or clinical claims (Section VIII).

II. RELATED WORK

A. Age-friendly mobile interface design

Systematic reviews of mobile design for older adults consistently identify text size, target size, icon interpretability, and navigation complexity as the dominant barriers. Awan *et al.* applied an analytical hierarchy process to prioritise usability barriers and ranked small font and screen size highest [7]. Elguera Paez and Zapata Del Río catalogue challenges across the literature [8]. Gomez-Hernandez *et al.* derive design guidelines through thematic synthesis [10], and Amouzadeh *et al.* provide a recent consolidation of age-friendly design recommendations [11]. Liu *et al.* review interface and persuasive feature design in mobile health applications for older adults [9]. Our elder surface is an implementation of this consensus; our contribution is not the guidelines but the demonstration that offloading configuration to a second surface is what makes conformance affordable.

B. Medication reminder and adherence systems

Poorcheraghi *et al.* report a randomised controlled trial of a mobile drug management application among older adults with polypharmacy, finding improved adherence and reduced hospital readmission [5]. Salmensuu *et al.* systematically review reminder technologies for home-dwelling older citizens [6]. Wiecek *et al.* report real-world engagement and adherence evidence for older adults [15]. These studies evaluate outcomes; they do not generally report whether reminders were delivered by the operating system as scheduled, which our Section III-D treats as a first-class engineering problem.

C. Dyadic and caregiver-mediated health systems

Quinn *et al.* conducted a dyad usability study of a mobile application used by older adults together with informal caregivers, observing that interaction was predominantly caregiver-mediated [12]. Benmessaoud *et al.* designed a dyadic digital health module for chronic disease shared care, identifying disease-invariant features for patient–caregiver dyads [13]. Nimmanterdwong *et al.* review human-centred design practice in mobile health for older adults [16]. This work establishes the dyad as the unit of design. We extend it architecturally: rather than one application used by two people, we build two surfaces of deliberately unequal complexity over one access-controlled record.

D. Personal emergency response systems

Personal emergency response systems (PERS) have a long history and a well-documented failure mode: Hamill *et al.* report call-centre estimates attributing as much as 85% of calls to false alarms, largely accidental activations [17]. This motivates our cancellable grace period. The complementary failure — an alarm the user believes was raised but which never reached anyone — is less discussed in that literature, and is what fail-honest alerting addresses.

E. Gap

Across these strands, interface legibility for the older adult and outcome efficacy are well covered; delivery honesty and deployability under platform policy are not. Few studies in the reviewed elder-care literature explicitly report a taxonomy of silent implementation failures, or treat application-store permission policy as an architectural design constraint on the emergency path.

III. SYSTEM DESIGN

A. Asymmetric dual-surface architecture

The system presents two Android surfaces backed by one record. The caregiver surface carries the full management vocabulary: profiles, medicines, schedules, reminders, contacts, vitals, adherence history, shortcuts, shared access, and an emergency monitor across seventeen navigation destinations. The elder surface carries seven: an alarm control, today's medicines, contacts, vitals entry, media shortcuts, a home screen, and a settings screen containing three controls, plus a one-time binding screen seen only at set-up. Contacts on the elder surface are *photo-first*: family members are presented as photographs rather than text labels, so recognition does not depend on reading a name.

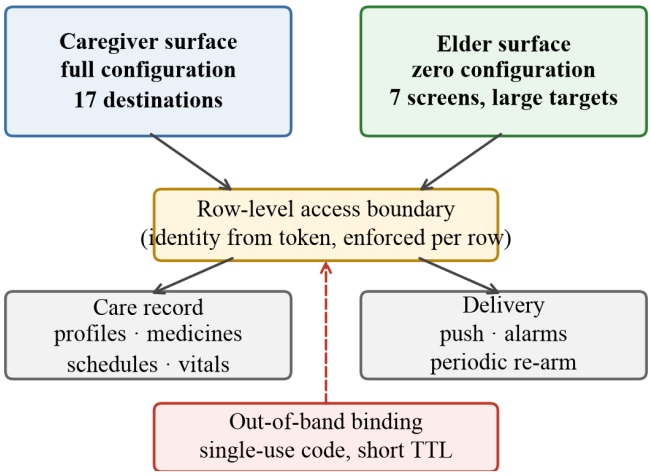


Fig. 1. System structure. Two clients of deliberately unequal complexity share one care record behind a per-row access boundary that derives identity from the caller's token. The out-of-band binding step (bottom) establishes which physical device belongs to which older adult, a link federated sign-in does not provide.

The asymmetry is the design. Because no configuration occurs on the elder surface, that surface has no forms to validate, no destructive actions to guard, no navigation deeper than two levels, and no state the older adult can corrupt. This is what makes generous type and target sizing affordable: a screen that must accommodate a scheduling form cannot also give every control 96 dp of height. Section V-A quantifies the resulting difference between the two surfaces.

Both surfaces authenticate with a federated identity provider and address the same record through a per-row access boundary. Thirty-three row-level policies across seventeen tables derive the

caller's identity from the verified token rather than from any client-supplied parameter, so a modified client cannot widen its own visibility. All privileged transitions — creating a profile, issuing a code, redeeming one, changing another member's access — are server-side routines rather than client writes.

B. Out-of-band device binding

Elder-care systems must answer a question ordinary applications do not: *which physical device belongs to the person being cared for?* The caregiver creates the profile, so the profile cannot identify its subject by who created it.

Our first implementation matched on phone number. The caregiver entered the older adult's number; when a device signed in with that number, the two were linked. This works under telephone-number-based authentication and fails completely under federated sign-in, which returns an email address and a provider subject identifier and no telephone number at all. After migrating to federated sign-in, the stored telephone number for every account was the empty string, and the matching predicate could never be satisfied. The failure was silent: links were recorded in a pending state, and the resolution routine compared an empty string against a real number indefinitely. Section V-C returns to this as an instance of a general hazard.

The repair is an out-of-band binding primitive. The caregiver requests a six-digit code bound to the profile; the code is single-use and expires after seven days. On the older adult's device, after federated sign-in, entering the code claims the profile for that account. The channel by which the code travels is deliberately unspecified — spoken aloud in the same room, sent over any messaging application, or typed by the caregiver on the older adult's device — because the caregiver is present in the overwhelming majority of set-ups [12].

Three properties matter. The code is *single-use*, so a shared code cannot be replayed. It is *short-lived*, bounding the window in which a guessed code is useful. And responses are *indistinguishable* between an unknown code and an expired one, so the endpoint cannot be used to enumerate which codes exist. A profile already bound to one account cannot be silently rebound to another. The same primitive, parameterised by an access level, later replaced the equally broken caregiver-invite path, which had depended on the same telephone-number assumption.

C. Fail-honest alerting

We adopt a single rule: *the interface never asserts an outcome it has not observed.* Applied to the emergency path, this produces the machine in Fig. 2.

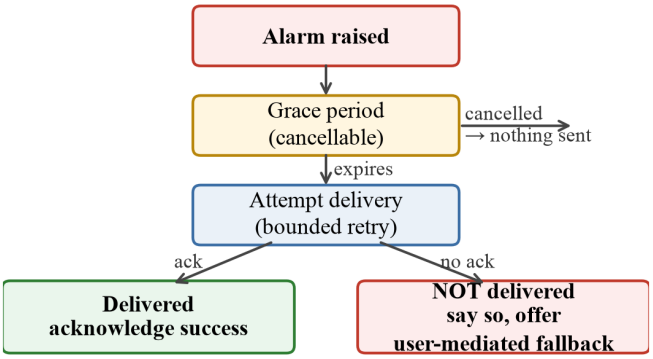


Fig. 2. The emergency path. A cancellable grace period addresses accidental activation, which dominates personal emergency response system traffic [17]. The terminal states are distinguished by observed delivery, not by having attempted it: an undelivered alarm says so and offers a user-mediated fallback.

Pressing the alarm control begins a five-second cancellable grace period. This is a direct response to the dominant PERS failure mode — Hamill *et al.* report estimates of up to 85% of call-centre traffic arising from accidental activation [17] — and it costs a genuine emergency five seconds.

If the period expires, the client attempts to record the alarm on the server, which is what reaches caregivers, using bounded retry with an increasing delay between attempts. The terminal state is then selected by whether that record was *acknowledged*, not by whether it was attempted. On acknowledgement, the interface confirms delivery. On failure it states plainly that the alarm was not delivered and presents two large controls: one opens the device's own messaging application with the emergency text and a location link pre-composed, and one dials the primary contact.

An earlier revision of this screen displayed a success confirmation unconditionally. It was possible for an older adult to be shown a green tick and the words *your family has been notified* when the emergency event had not in fact reached the caregiver-facing system. We regard this class of defect as the most serious a care system can contain, because it converts a recoverable failure into an unrecoverable one: a user who believes help is coming stops seeking it.

D. Redundant reminder arming

Medication reminders must fire on a device the application may not be running on, under an operating system actively suppressing background work. Android's Doze and App Standby defer alarms, jobs, and network access during idle, and manufacturer power management layers are typically more aggressive [14].

An early revision armed the day's alarms only when the older adult opened the application. The consequence is easy to state and was easy to miss: an older adult who did not open the application received no reminders that day — precisely the user for whom the reminders exist. We now arm through three independent paths: on application open; from a periodic background worker that re-arms the current day every four hours and skips doses already answered; and from a boot-completed receiver that re-arms from a persisted set, since alarms do not survive a restart.

Exact alarms are requested where the platform grants them for medication reminders, with a windowed fallback when the permission is unavailable.

Two correctness properties proved necessary in implementation. Alarm keys must be namespaced by source, because the key's hash is used as both the pending-intent request code and the notification identifier; an un-namespaced key allowed a general reminder and a medication dose to silently replace one another. And answering a dose must cancel its pending alarm, or an older adult who takes a tablet early is told to take it again at the scheduled time — a prompt toward double-dosing generated by an adherence tool.

E. Permission-minimal degradation

Emergency features in the research literature routinely assume the application may send a short message directly. Current Google Play policy substantially restricts this: messaging and call-log permissions are limited to applications whose primary function is messaging or telephony, and a care application declaring them is likely to face review or rejection [18], [19].

We therefore removed direct message-sending and direct dialling entirely. The emergency path reaches caregivers over the network; where a message is warranted, the application hands a pre-composed message to the device's own messaging application through a standard intent, and the user presses send. This costs one tap in the degraded path and removes two restricted permissions, reducing the declared set to eleven, none of which is restricted. We report this because deployability is a research-relevant property: a prototype that cannot be distributed cannot be evaluated at scale, and this constraint is rarely stated in the elder-care literature.

F. Localisation and perceptual settings

The elder surface is localised into English, Hindi, Marathi, and Gujarati, and offers three type scales and a high-contrast mode that composes with the operating system's own accessibility scaling rather than overriding it. Language matters here beyond convenience: interfaces presented only in English are a documented barrier for older adults in India, particularly older women with lower English literacy [3], [4].

IV. IMPLEMENTATION

Both clients are native Android applications written in Kotlin using Jetpack Compose, targeting API level 36 with a minimum of API 24. Persistence and access control are provided by Supabase-hosted PostgreSQL with row-level security, reached over its generated REST interface; privileged operations are database routines executing with definer rights. Push delivery uses Firebase Cloud Messaging with data-only messages, since notification-payload messages bypass the application's own handler when the application is backgrounded and would prevent the client from applying its own delivery rules — including the rule that emergency alerts are exempt from the application's general notification preference.

TABLE I

Implementation size, measured from the repository

Component	Files	Lines
Caregiver surface	27	3,942
Elder surface	6	1,271
Data access and models	12	1,082
Reminder and alarm engine	4	329
Notification and push	2	147
Authentication	1	79
Application total (Kotlin)	83	13,473
Automated tests (Kotlin)	26	3,010
Database migrations (SQL)	15	1,121

The test suite comprises 142 automated tests executing on the Java virtual machine with a shadowed Android runtime, requiring no device or emulator. Coverage is concentrated where silent failure is most likely: alarm scheduling and cancellation, offline durability of adherence records, delivery-contract behaviour of the push path, translation-table integrity across the four languages, and regression tests for each defect in Section V-C. Static analysis reports no errors at release configuration.

V. EVALUATION

We evaluate three properties measurable from the artefact: conformance of the elder surface to age-friendly sizing guidance, interaction cost for core tasks, and the defect profile of the feature-complete system. All figures in this section are generated by a script that parses the application source directly, so they cannot drift from the implementation.

A. Conformance of the elder surface

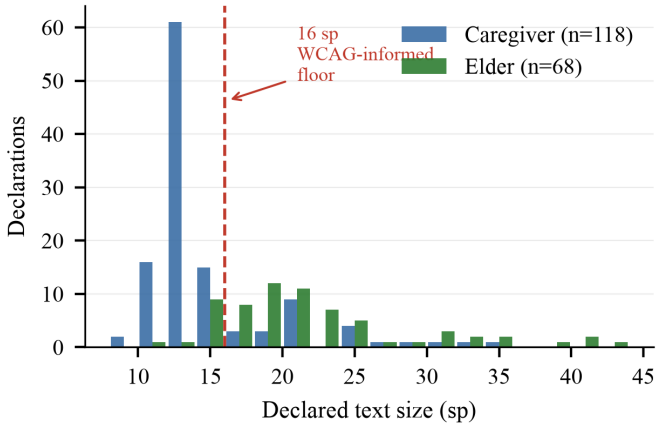


Fig. 3. Distribution of declared text sizes across the two surfaces, parsed from source. The caregiver surface concentrates at 12–13 sp; the elder surface is shifted upward with a long tail to 72 sp for the alarm control. Medians are 13 sp and 20 sp respectively.

Fig. 3 shows every declared text size on each surface. The elder surface has a median of 20 sp against 13 sp for the caregiver surface, and 16.4% of its declarations fall below 16 sp against 79.7% on the caregiver surface. The residual small sizes on the elder surface are supporting captions rather than primary content. Because the elder surface additionally composes a user-selected

scale of 1.0, 1.15, or 1.3 with the operating system's own font scaling, the effective floor rises further when either is increased.

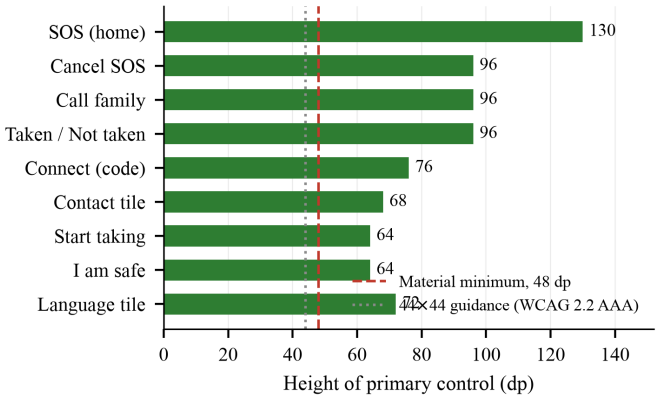


Fig. 4. Height of the primary control on each elder screen, parsed from source. Every primary control exceeds the 48 dp platform minimum and the widely cited 44×44 target-size recommendation; the median is 76 dp and the emergency control is 130 dp.

Fig. 4 reports the height of the primary control on each elder screen. The smallest is 64 dp, one third above the 48 dp platform minimum [20] and comfortably above the widely cited 44×44 target-size recommendation associated with WCAG 2.2 AAA [21], noting that WCAG is specified in CSS pixels rather than Android density-independent pixels, so the comparison is indicative rather than exact; the median is 76 dp. The emergency control is 130 dp, roughly 2.7 times the platform minimum, reflecting its priority. We report this as conformance evidence, not as a usability result: meeting a sizing guideline is necessary, not sufficient, and only a study with older adults can establish sufficiency.

B. Task interaction cost

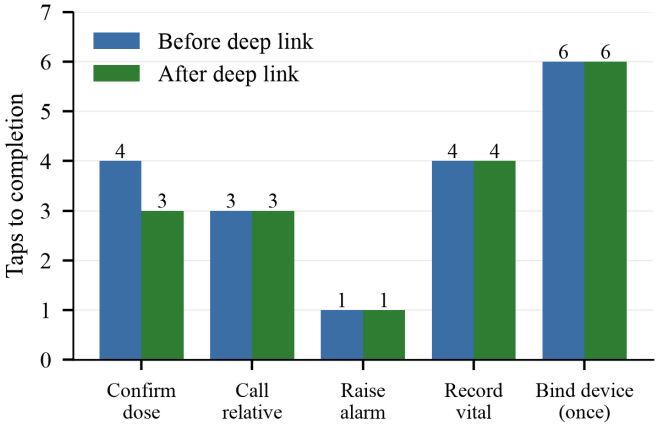


Fig. 5. Taps to completion for core tasks on the elder surface, counted from the notification or from the home screen. Deep-linking the medication reminder removed one tap from the most frequent task; other paths were already minimal.

Fig. 5 counts taps to completion for the elder surface's core tasks. Raising an alarm costs one tap. Confirming a dose from its reminder costs three, following a change that routes the notification directly to the medication screen rather than to the home screen; previously it cost four. The saving is one tap, which sounds trivial and is not: each intermediate screen is an opportunity for an older adult to lose the thread, and an unconfirmed dose is recorded as missed and shown to the caregiver as a possible non-adherence event. Interaction cost on this surface is therefore coupled to data quality, not only to convenience.

C. Defect study

After the system was feature-complete and had been exercised on physical devices, we conducted a structured audit: a line-by-line review of all 83 source files together with targeted inspection of the database schema and access policies. We recorded every defect found and classified each by whether a user could have detected it from the interface alone. Every defect reported here was subsequently corrected, and each correction is traceable to a commit in the project's version-control history together with the regression test added for it.



Fig. 6. Defects found in the audit window, by class. Eight of thirteen (62%, red) would have failed with no user-visible signal: the interface either reported success or reported nothing at all.

Thirteen distinct defects were found. Fig. 6 gives the distribution. The dominant finding is that 8 of 13 (62%) were *silent*: the operation failed, or never occurred, while the interface reported success or reported nothing.

Three classes deserve comment.

- **Unreachable features (3).** Two complete features — general reminders for hydration and activity, and the caregiver invitation path — were written, exercised by the caregiver, stored correctly in the database, and delivered to nobody. General reminders were never armed as alarms on the older adult's device and no server job scanned for them. The invitation path depended on the telephone-number assumption broken by federated sign-in (Section III-B). In both cases the caregiver's interface behaved as though the feature worked. Both were subsequently repaired: general reminders are now armed through the same scheduling path as medication doses, and the invitation path was replaced by the binding primitive of Section III-B.
- **Lost state (3).** Three view models rebuilt their state object when a background refresh completed, discarding a transient completion flag that the screen used to navigate. A save that had already succeeded on the server therefore appeared to do nothing. The defect is timing-dependent, which is why it survived manual testing: it reproduces only when the refresh completes after the save, most likely on a slow connection.
- **Null failure messages (1 class, 17 sites).** Every repository write terminates in taking the first element of the server's response. When access policy hides the returned row, that list is

empty and the resulting exception carries no message. Seventeen call sites propagated that null message directly to the interface, rendering an empty error and presenting as an unresponsive button.

Two observations generalise beyond this system. First, *identity-model migration produces latent dead features*. Moving from telephone-number authentication to federated sign-in silently invalidated every code path that used a telephone number as a join key. Nothing failed loudly; two features simply stopped being reachable. Any system migrating to federated identity should audit its non-authentication uses of the prior identifier.

Second, *test suites do not detect unreachable features*. Our 142 tests passed throughout the period in which two features delivered nothing, because each tested component behaved correctly in isolation. The defect lay in the absence of a caller. Detection required tracing each data type from producer to consumer — a check we recommend as routine practice and for which we found no substitute.

D. Deployability

The application declares eleven permissions, none of them in a restricted group. Two restricted permissions present in an earlier revision were removed in favour of the user-mediated path of Section III-E. The release artefact is 20 MB and targets the currently required API level. The permission model was designed to remain compatible with current Google Play policy; the application has not been published to the store. We report this because a prototype that cannot be distributed cannot be studied longitudinally in the field.

E. Threats to validity

The defect study has a single observer and no inter-rater agreement, so the classification carries the authors' judgement. The audit window covers the period after feature completion, not the whole development history, so the reported count is not a defect density for the project. Conformance measurements are static properties of the source and say nothing about perception or comprehension. Tap counts are analytic, counted by the authors on the implemented flows, and are not observed task times.

VI. DISCUSSION

A. Silence is the dominant risk

The audit's central result is that most defects in a feature-complete care application were invisible from the interface. This has a specific consequence for elder care that it does not have for, say, a photo application. The older adult often cannot diagnose a silent failure, and the caregiver — the person who could — is looking at a different device that is also reporting success. A missing reminder produces no artefact at all; its absence is indistinguishable from a day with no medication due. The system's own adherence record, which the caregiver trusts, will attribute the gap to the older adult.

We therefore propose that care systems adopt an explicit honesty discipline: acknowledge only observed outcomes; give every failure path a message a lay reader can act on; and, for any feature whose value is delivery, test the delivery rather than the storage.

B. Asymmetry as an accessibility budget

The measured gap between the two surfaces (Section V-A) is not the result of applying guidelines harder on one screen. It is a consequence of removing work from that screen. Type size and target size compete with information density; a surface that must present a scheduling form cannot also give every control 96 dp. By moving all configuration to the caregiver, the elder surface gains an accessibility budget it can spend on sizing and shallow navigation. We suggest this reframing — asymmetry as a budget rather than as simplification — is the transferable idea.

C. Policy as a design input

Application-store permission policy determined the architecture of our emergency path. This is unusual to state in a research paper and, we think, ought not to be. A design that cannot be distributed cannot accumulate the longitudinal evidence the field needs. Treating policy as a constraint at design time, rather than discovering it at submission time, is a practical contribution of this work.

VII. SCALABILITY CONSIDERATIONS AND FUTURE DEPLOYMENT

The architecture is designed to support independent care circles: circles are disjoint, queries are scoped to a profile, and no routine operation requires a cross-tenant scan, so growth in the number of circles should be horizontal. The heaviest recurring server task is a scheduled scan for unanswered doses, whose cost is proportional to active profiles with due doses rather than to total users. Large-scale performance has not been benchmarked; the following is architectural analysis rather than measurement.

Three properties keep per-user cost low. Reminder scheduling is performed on the device, so the server is not a timing authority and carries no per-dose timer. Adherence writes are small, append-only, and queue locally when offline. Media is limited to a few profile and medication photographs per circle.

The identified scaling risks are ordinary rather than architectural: push-token churn requires re-synchronisation on session restore, which we implement; a single scheduled scan is a single point of failure and would need partitioning; and per-row policy evaluation grows with policy count, currently thirty-three across seventeen tables, which should be benchmarked before any large deployment.

Path to evaluation. The immediate next step is not a feature. It is a study. We plan a dyad field deployment with older adults and their caregivers, measuring delivery-level outcomes the literature usually assumes: proportion of scheduled reminders actually presented by the device, time from alarm activation to caregiver acknowledgement, and frequency of the degraded emergency path. These are precisely the quantities our fail-honest instrumentation can record truthfully, which is the point of building it that way.

VIII. LIMITATIONS

We state these plainly because the paper's credibility depends on it.

- **No user study has been conducted.** No older adult or caregiver outside the development team has used the system under observation. We make no usability, adherence, or clinical claim. Every conformance number in Section V is a static property of the implementation.
- **No field measurement of delivery.** Reminder delivery has been verified on physical devices anecdotally and in a shadowed runtime exhaustively, but we have no distribution of delivery latency across manufacturers and power-management regimes — the measurement that would substantiate Section III-D.
- **Single system, single team.** The defect taxonomy comes from one application audited by its own developers. It is a case study, not a population estimate.
- **Localisation is untested with native speakers.** The four language tables are complete and machine-checked for structural integrity, but have not been reviewed by native speakers for register or clarity for older readers.
- **The system is not a medical device** and provides no diagnostic or dosing advice. Vital-sign banding is presented as general reference information only.

IX. CONCLUSION

We presented CareCompanion, a caregiver-mediated elder-care system built on three commitments: an asymmetric dual-surface architecture that buys the older adult an accessibility budget by removing configuration from their device; out-of-band device binding that repairs an identity gap federated sign-in introduces; and fail-honest alerting, under which the interface never asserts an outcome it has not observed.

Our measurements show the asymmetry is real — a median 20 sp against 13 sp, every primary control at or above 64 dp — and our audit shows why the honesty discipline matters: 62% of the defects in a feature-complete system would have failed without any user-visible signal, including two features stored faithfully and delivered to nobody. These results suggest that elder-care systems should be evaluated not only for accessibility and functionality, but also for whether users receive truthful feedback when critical operations fail. Future work will evaluate CareCompanion with older adults and caregivers in real-world use.

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